

Annotating MAVE results with the Ensembl Variant Effect Predictor



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**Multiplex assays of variant effects (MAVEs):
Approaches, Analysis and Interpretation - 2025/11**



MAVE data assessment

You have scores for large numbers of sequence changes you MAVE assay - how do your results correlate with the available knowledge about the bases assayed?

There are useful reference datasets available for results checking

Here: describe available resources, and how to integrate them using the Ensembl Variant Effect Predictor



Outline

- Using Ensembl VEP
- Predicting the effect of sequences changes on functional genomic elements
- Using population data to assess sensitivity to change
- Known variant impact
- Variant scoring tools
- How it works
- Example command line

Ensembl VEP - web interface

<https://www.ensembl.org/Tools/VEP>

- Multiple input formats supported
- Maximum job: 50M file
 - equivalent to ~2 million lines in VCF
- Analysis jobs managed by scheduling system
- Simple to use
- Least configurable option

Variant Effect Predictor

New job

Clear form

Species:

Human (Homo sapiens)

Assembly: GRCh38.p12 (If you are looking for VEP for Human GRCh37, please go to [GRCh37 website](#).)

Name for this job (optional):

Input data:

Either paste data:

1 182712 182712 A/C 1

Run instant VEP for current line

Examples: [Ensembl default VCF](#), [Variant identifiers](#), [HGVIS notations](#)

Or upload file:

Browse...

No file selected.

Or provide file URL:

Transcript database to use:

☒ Ensembl/GENCODE transcripts

☐ Ensembl/GENCODE basic transcripts

☐ RefSeq transcripts

☐ Ensembl/GENCODE and RefSeq transcripts

Additional configurations:

Identifiers Additional identifiers for genes, transcripts and variants

Variants and frequency data Co-located variants and frequency data

Additional annotations Additional transcript, protein and regulatory annotations

Predictions Variant predictions, e.g. SIFT, PolyPhen

Filtering options Pre-filter results by frequency or consequence type

Run

Recent jobs

Ensembl VEP - web interface

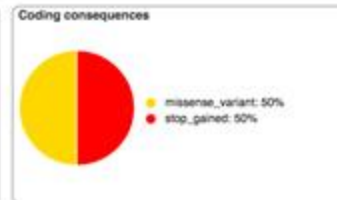
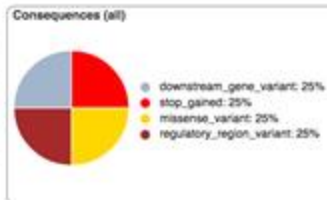
Summary information

Variant Effect Predictor results

Job details

Summary statistics

Category	Count
Variants processed	2
Variants filtered out	0
Novel / existing variants	0 (0.0) / 2 (100.0)
Overlapped genes	3
Overlapped transcripts	3
Overlapped regulatory features	1



Results preview

Navigation (per variant)

Filters

Uploaded variant

its

defined

Download

All: VCF VEP TXT

Variant Effect Predictor - PDB

Variant var2 (miss. gene)

Select PDB file (model): - 1-142 - Coverage | PDBs: 1-142 | ENSP: 1-142 | 100% of ENSP length



3D Viewer Help | Reset view

Ensembl-PDBs mapping

Label: PDB | ENSP

Coverage: 1-142 | 1-142

Exons

Exons: 1-142

Protein Information

Pfam: 1-142

Gene3D: 1-142

Variants

Variant var2

ID: 25 | 25

var2: 25 | 25

Altered/missing sequence: from 25 | from 25 | go

SIFT: 100 | 100

PolyPhen: 100 | 100

Consequence	Symbol	Feature	CDS position	Protein position	Amino acids	Existing variant	SIFT	PolyPhen	Domains	AF	Clinical significance
missense_variant	AGT	ENST00000296667	803	268	M/T	rs699, CM920010	1	0	15 Protein domains	0.7051	benign, risk_factor
downstream_gene_variant	AL512328.1	ENST0000412344	-	-	-	rs699, CM920010	-	-	-	0.7051	benign, risk_factor
stop_gained	CLC	ENST00000221804	73	25	R/*	rs761258843, G05M5777949	-	-	24 Protein domains	-	-
regulatory_region_variant	-	ENSP00000269246	-	-	-	rs761258843, G05M5777949	-	-	-	-	-

Ensembl VEP - REST interface

VEP

Resource	Description
GET vep/:species/hgvs/:hgvs_notation	Fetch variant consequences based on a HGVS notation
POST vep/:species/hgvs	Fetch variant consequences for multiple HGVS notations
GET vep/:species/id/:id	Fetch variant consequences based on a variant identifier
POST vep/:species/id	Fetch variant consequences for multiple ids
GET vep/:species/region/:region/:allele/	Fetch variant consequences
POST vep/:species/region	Fetch variant consequences for multiple regions

<https://rest.ensembl.org/>

- language-agnostic programmatic access
- ideal for:
 - integrating information on a single variant into a webpage
 - integration into small scale annotation pipelines
- limits: 55,000 requests per hour (~15 per second)

Optional

Name	Type	Description	Default
Biosum62	Boolean	Include BLOSUM62 amino acid conservation score (plugin details)	0
Conservation	Boolean	Retrieves a conservation score from the Ensembl Compara databases for variant positions (plugin details)	0
GeneSplicer	Boolean	Detects splice sites in genomic DNA (plugin details)	0
MaxEntScan	Boolean	Sequence motif and maximum entropy based splice site consensus predictions (plugin details)	0
Phenotypes	Boolean	Retrieves overlapping phenotype information (plugin details)	0
dbNSFP	String	Include fields from dbNSFP, a database of pathogenicity predictions for missense variants. Multiple fields should be separated by commas. See dbNSFP README for field list. (plugin details)	Not used
dbcsnv	Boolean	Predictions for splicing variants from dbcsnv. (plugin details)	0
distance	Integer	Change the distance to transcript for which VEP assigns upstream and downstream consequences	5000
domains	Boolean	Include names of overlapping protein domains	0
failed	Boolean	When checking for co-located variants, by default variants flagged as failed by Ensembl's QC pipeline will be excluded. Set this flag to 1 to include such variants	0
hgvs	Boolean	Include HGVS nomenclature based on Ensembl stable identifiers	0
merged	Boolean	Use merged Ensembl and RefSeq transcript set to report consequences (human only)	0
miRNA	Boolean	Determines where in the secondary structure of a miRNA a variant falls (plugin details)	0
minimal	Boolean	Convert alleles to their most minimal representation	0

Ensembl VEP - command line interface

- Fastest option
- Total data privacy
- Reference data downloadable in compact data files (caches)
- Docker image + VM supported
- Nextflow workflow available
- Highly customisable in both reference data used and functionality applied (via plugins)



<https://github.com/Ensembl/ensembl-vep/>
<https://hub.docker.com/r/ensemblorg/ensembl-vep>

Variant input formats supported

Ensembl VEP is optimized to annotate variants described in Variant Call Format (VCF) and ordered by genomic location

Identifiers from a range of popular resources can be used as input for on-line analysis:



HGVS at genomic, transcript and protein level is supported, though protein level descriptions may give ambiguous results

NCBI SPDI format is supported at genomic level

A simple position-based tab delimited format is supported

Variant filtering

Ensembl VEP includes utilities to filter results using customisable criteria

filter_vep

--ontology --filter "Consequence is coding_sequence_variant"

--filter "AF < 0.01 or not AF"

--filter "CADD > 24"

Available as a
command line utility or
via the Ensembl VEP
web interface

Results preview

Navigation (per variant)

Page: 1 of 1 | Show: 1 variants

Filters

MANE SELECT is defined

CADD PHRED > 24

Clear filters

Match all of the above rules

Update

Uploaded variant

is defined

Add

Download

New job

All: VCF VEP TXT

Filtered: VCF VEP TXT

BioMart: Variants Genes

Show/hide columns (71 hidden)

Location	Consequence	Symbol	Feature	Codons	Existing variant	SWISSPROT	Protein matches	gnomADg AF	gnomADg FIN AF	Clinical significance	EVE CLASS	MaveDB protein change	MaveDB score	MaveDB URN
16:1314031-1314031	start_lost	UBE2I	ENST00000397514.8	ATG/TTG	-	P63279.198	Protein Structure View 27 Protein matches	-	-	-	-	p.Met1Leu	0.4779	urn:mave:db:00000001-8-1
1:9975672-9975672	missense_variant	NMNAT1	ENST00000377205.6	CGG/TGG	rs763325435, Q9HAN9.187 CM127767		Protein Structure View AlphaFold model 29 Protein matches	1.972e-05	9.443e-05	pathogenic	Pathogenic	-	-	-

Outline

- Using Ensembl VEP



Predicting the effect of sequences changes on functional genomic elements

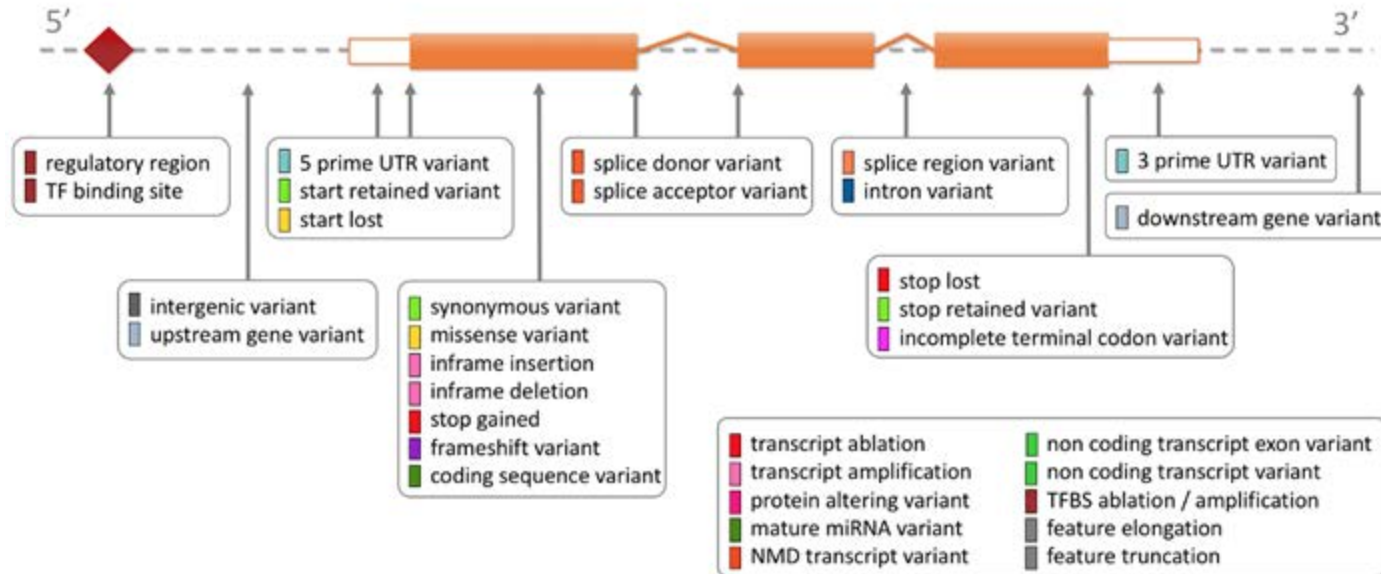
- Using population data to assess sensitivity to change
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Molecular consequence prediction

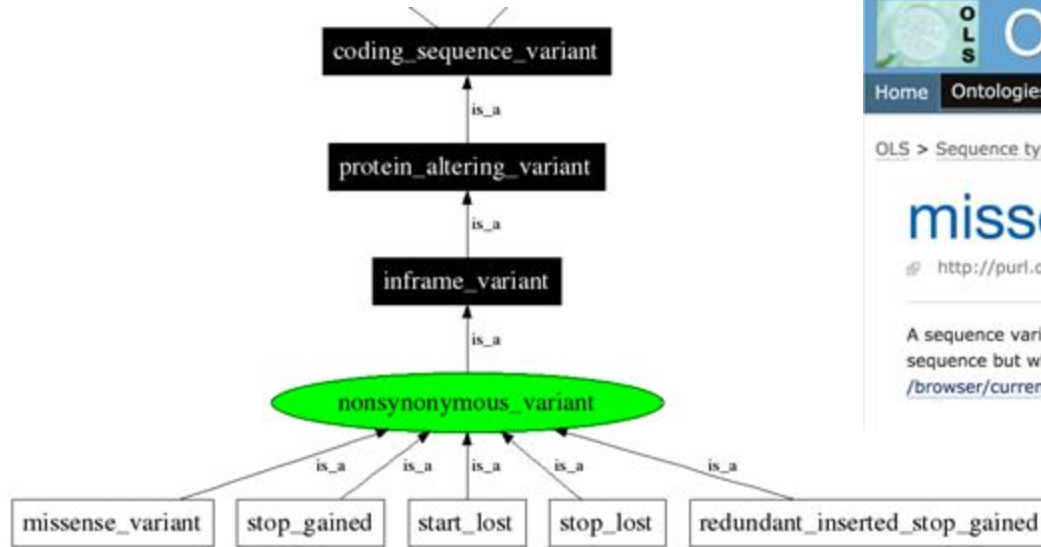
Variants are mapped to transcript structures

The location type is identified

The expected molecular consequence of the change is predicted



Sequence Ontology

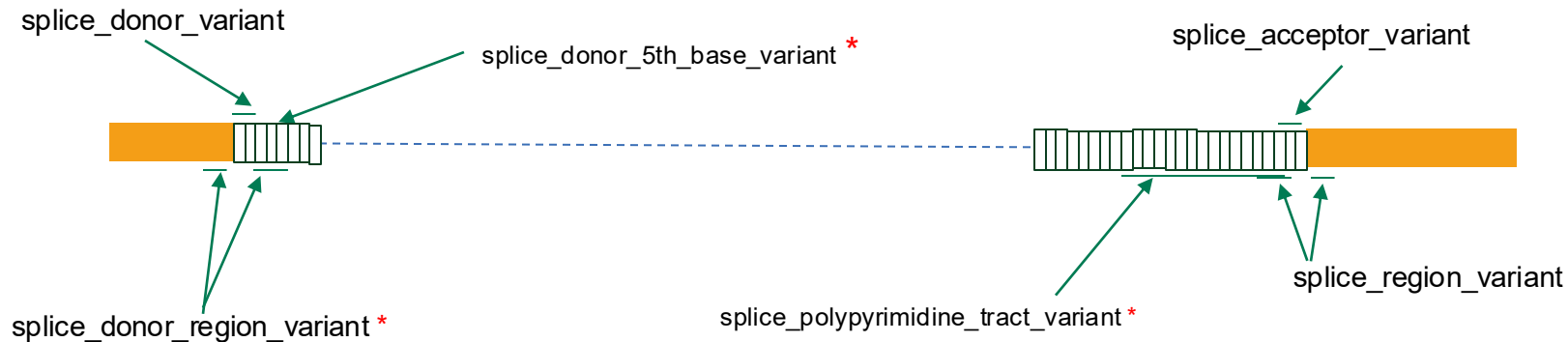


The screenshot shows the Ontology Lookup Service (OLS) interface. The header includes the OLS logo and the title "Ontology Lookup Service". Below the header are navigation links: Home, Ontologies, Documentation, About, and Contact Us. The main content area displays the term **missense_variant** from the Sequence types and features ontology (SO). The term is highlighted in a blue box. Below the term is its URI: http://purl.obolibrary.org/obo/SO_0001583. A search bar is visible on the right. The definition of the term is provided: "A sequence variant, that changes one or more bases, resulting in a different amino acid sequence but where the length is preserved. [EBI:gr EBI:fc http://www.sequenceontology.org/browser/current_svn/term/SO:ke]".

Standards enable understanding and comparisons across different resources

The use of Sequence Ontology terms to describe a variants predicted molecular consequence has been agreed by many of the major variant databases and tool providers including Ensembl, UCSC Genome Browser, dbSNP, SnpEff

Variants in/near splice sites



'Splice site variant' is not reported – more specific terms are used

More detailed terms (*) added due to Lord et al 2019 - Pathogenicity and selective constraint on variation near splice sites (10.1101/gr.238444.118)

Gene annotation – transcript choice

Two groups create gene annotation for multiple species

- Both use a variety of evidence to determine transcript structures
- Tissue specific expression → many different splice isoforms

Many species have community-derived gene annotation

Transcript choice has a large impact on variant annotation

- Both groups provide default transcripts but considering these alone will miss information
- Best approach – annotate variants against a comprehensive set of transcripts and consider transcript quality information when determining variant impact.



McCarthy DJ,, et al. Choice of transcripts and software has a large effect on variant annotation.
Genome Med. 2014;6:26.

Human transcript sets

GENCODE Primary:

- Contains nearly all human exons in the minimum number of transcripts, limiting the number of transcripts to be considered without losing information

MANE - Matched Annotation from NCBI and EMBL-EBI:

- A set of standard default transcripts for variant reporting in protein coding genes
- Matches the GRCh38 assembly
- Identical between NCBI RefSeq and Ensembl/GENCODE



Two tiers :

- MANE Select - one per gene, representative of biology at each locus; well-supported, expressed, conserved
- MANE Plus Clinical - alternate transcripts to capture additional clinically reportable variants

A joint NCBI and EMBL-EBI transcript set for clinical genomics and research.

Morales J, et al. Nature. 2022 Apr;604(7905):310-315. DOI: 10.1038/s41586-022-04558-8.

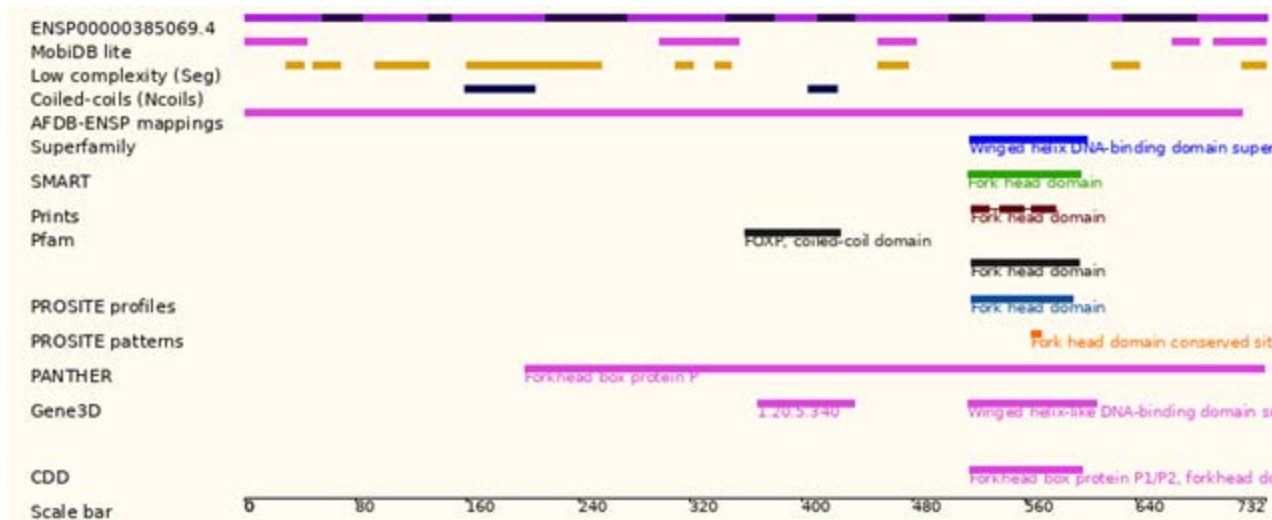
Transcript data in the human Ensembl VEP cache

Transcript quality information

- Transcript Support Level (TSL)
- APPRIS

Protein mappings

- UniProt
- PDBe
- protein families, domains and sites from InterProScan



InterProScan results for an Ensembl protein sequences.

Regulatory regions



Open chromatin

Histone modification assays

Transcription factor
binding assays and motifs

Ensembl Regulatory Build

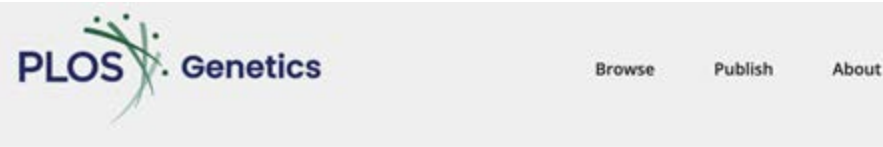
- promoters
- enhancers
- transcription factor binding sites

Consider potential impact
on gene expression

Uploaded variant	Consequence	Feature	Biotype	Motif name	Motif position	High info position	Motif score change
rs1374682526	regulatory_region_variant	ENSR00000060894	promoter	-	-	-	-
rs1374682526	TF_binding_site_variant	ENSM00155148274	-	ENSM00155148274	10	Y	-0.093
rs1374682526	TF_binding_site_variant	ENSM00000058901	-	ENSM00000058901	13	Y	-0.134
rs1374682526	TF_binding_site_variant	ENSM00000121206	-	ENSM00000121206	10	Y	-0.091

Non-canonical open reading frame analysis

There is increasing interest in open reading frames within long noncoding RNAs and in the presumed untranslated region of protein coding genes



OPEN ACCESS

REVIEW

Gene Expression Regulation by Upstream Open Reading Frames and Human Disease

Cristina Barbosa, Isabel Peixeiro, Luísa Romão

Published: August 8, 2013 • <https://doi.org/10.1371/journal.pgen.1003529>

A comprehensive catalog of predicted functional upstream open reading frames in humans

Patrick McGillivray, Russell Ault, Mayur Pawashe, Robert Kitchen, Suganthi Balasubramanian, Mark Gerstein, Author Notes

Nucleic Acids Research, Volume 46, Issue 7, 20 April 2018, Pages 3326–3338, <https://doi.org/10.1093/nar/gky188>

Standardized annotation of translated open reading frames

[Jonathan M. Mudge](#), [Jorge Ruiz-Orera](#), [John R. Prensner](#), [Marie A. Brunet](#), [Ferriol Calvet](#), [Irwin Jungreis](#), [Jose Manuel Gonzalez](#), [Michele Magrane](#), [Thomas F. Martinez](#), [Jana Felicitas Schulz](#), [Yucheng T. Yang](#), [M. Mar Albà](#), [Julie L. Aspden](#), [Pavel V. Baranov](#), [Ariel A. Bazzini](#), [Elspeth Bruford](#), [Maria Jesus Martin](#), [Lorenzo Calviello](#), [Anne-Ruxandra Carvunis](#), [Jin Chen](#), [Juan Pablo Couso](#), [Eric W. Deutsch](#), [Paul Flicek](#), [Adam Frankish](#), ... [Sebastiaan van Heesch](#) + Show authors

Nature Biotechnology 40, 994–999 (2022) | [Cite this article](#)

Efforts are underway to create catalogues of translated non-canonical ORF, assayed using Ribosome profiling

Non-canonical open reading frame analysis

RiboseqORFs plugin - Predicts the molecular consequences of a short variant with respect to open reading frames discovered in noncoding transcripts/regions



UTRAnnotator plugin - annotates variants in 5' untranslated regions (5'UTR) that create or disrupt upstream open reading frames. (Zhang et al 2021)

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Using population data to assess sensitivity to change

- Known variant impact
- Variant scoring tools
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Allele frequency reference sets

Comparison of the MAVE score for an allele to its population frequency is useful - if an allele is common in a large set of healthy individuals, a severely deleterious MAVE score is less likely

Set	Individuals
1000 Genomes Project	2,504
TOPMed (heart, lung, blood, and sleep disorders)	62,784
gnomAD genomes	76,215 (v4)
gnomAD exomes	730,947 (v4)
ALFA (dbGaP - includes cases)	204,108 (v3)
AllofUs	633,540 (v7.1)

- Be aware of cases in the allele frequency data sets
- If the gene you are assaying is associated with a fairly common disease, consider this on applying frequency checks

Ensembl VEP human caches contain:



Other datasets can be added as custom annotations

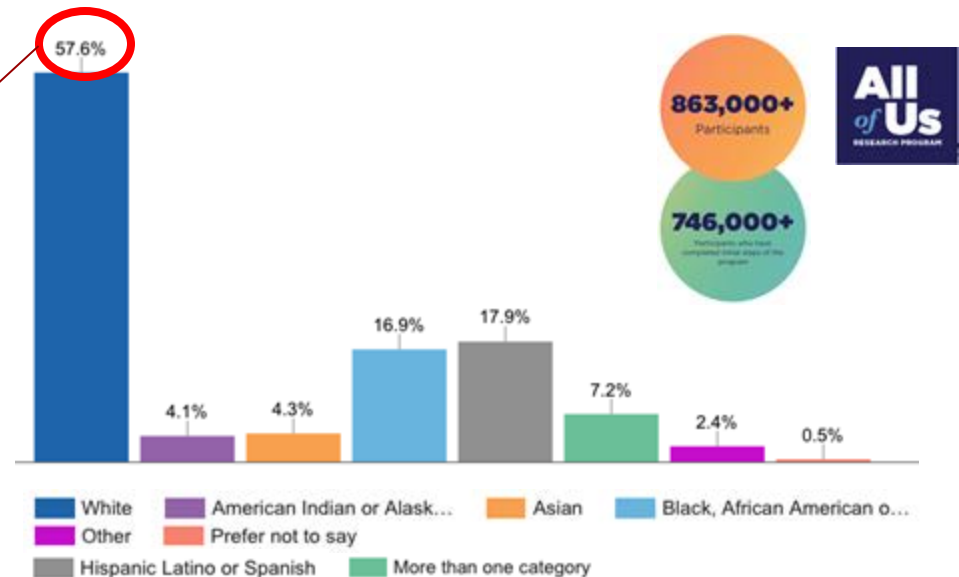
Allele frequency reference sets

Current large population frequency datasets are biased towards people of European ancestry.

This causes problems for :

- personalised medicine in other populations
- filtering 'common' variants in rare disease investigations
- bias in pathogenicity predictors trained on population frequencies

gnomAD v4*			
	Sample count	%	Increase from v2
Admixed American	30,019	3.72%	1.7x
African/African American	37,545	4.65%	3x
Ashkenazi Jewish	14,804	1.83%	2.9x
East Asian	22,448	2.78%	2.3x
European^	622,057	77.07%	8.1x
Middle Eastern	3,031	0.38%	19.2x
Remaining^	31,712	3.93%	8.8x
South Asian	45,546	5.64%	3x
Total	807,162	-	-



Tolerance to change - conservation

Homo_sapiens	ACTGGCTCCC AT CAGGGAGCA
Pan_paniscus	ACTGGCTCCC GT CAGGGAGCA
Pan_troglodytes	ACTGGCTCCC GT CAGGGAGCA
Gorilla_gorilla	ACTGGCTCCC GT CAGGGAGCA
Pongo_abelii	ACTGGCTCCC GT CAGGGAGCA
Nomascus_leucogenys	ACTGGCTCCC GT CAGGGAGCA
Cercocebus_atys	ACTGGCTCCC GT CAGGGAGCA
Mandrillus_leucophaeus	ACTGGCTCCC GT CAGGGAGCA
Papio_anubis	ACTGGCTCCC GT CAGGGAGCA
Macaca_fascicularis	ACTGGCTCCC GT CAGGGAGCA
Macaca_mulatta	ACTGGCTCCC GT CAGGGAGCA
Macaca_nemestrina	ACTGGCTCCC GT CAGGGAGCA
Chlorocebus_sabaeus	ACTGGCTCCC GT CAGGGAGCA
Rhinopithecus_bieti	ACTGGCTCCC GT CAGGGAGCA
Rhinopithecus_roxellana	ACTGGCTCCC GT CAGGGAGCA
Aotus_nancymae	ACTGGCTCCC GT CAGGGAGCA
Callithrix_jacchus	ACTGGCTCCC GT CAGGGAGCA
Cebus_imitator	ACTGGCTCCC GT CAGGGAGCA
Saimiri_boliviensis_boliviensis	ACTGGCTCCC GT CAGGGAGCA
Carlito_syrichta	ACTGGCTCCC GT CAGGGAGCA
Microcebus_murinus	ACTGGCTCCC GT CAGGGAGCA
Prolemur_simus	ACTGGCTCCC GT CAGGGAGCA
Propithecus_coquereli	ACTGGCTCCC GT CAGGGAGCA
Otolemur_garnettii	ACTGGCTCCC GT CAGGGAGCA

Conservation measures reveal regions which are highly similar **across** species.

- calculated from the genome wide alignment of the sequenced genomes of many different species
- measures include GERP, phyloP

1 230710048 A -> G

GERP score: -3.69 (neutral)
restores 'ancestral allele'

Available as **Conservation** plugin or custom annotation

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- **Known variant impact**
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Variant disease and phenotype associations

Multiple resources provide phenotype/ disease associations to variants

- Data is highly heterogeneous
- Important to consider by allele



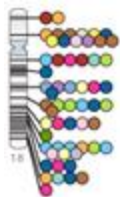
ClinVar

Assertions of likely pathogenicity for germline variants

Assertions of likely oncogenicity for somatic variants

Confidence ratings for assertions

Updated monthly



NHGRI- EBI
GWAS
Catalog

Statistical measures of association with a trait/ disease from large scale genome wide case- control studies

Details available via Ensembl VEP **Phenotype** plugin

Experimental: Clinvar annotation mapping via **Paralogs** plugin

Functional data - interactions



IntAct shares molecular interaction data, curated from reports of assays in the scientific literature.

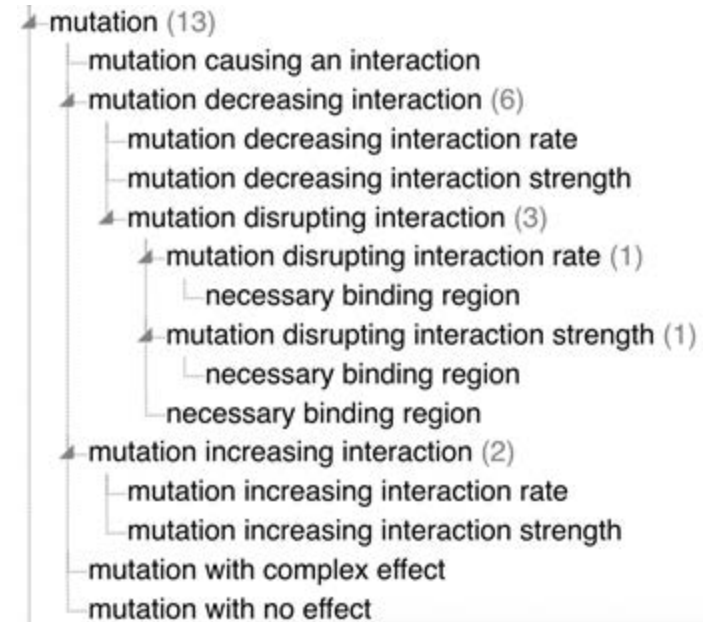
>72,000 records describing the effect of protein changes in human on interactions

Information described in standard formats with links to publication providing detail

<https://www.ebi.ac.uk/intact/home>

IntAct plugin and genome-mapped data available

Molecular Interactions Controlled Vocabulary



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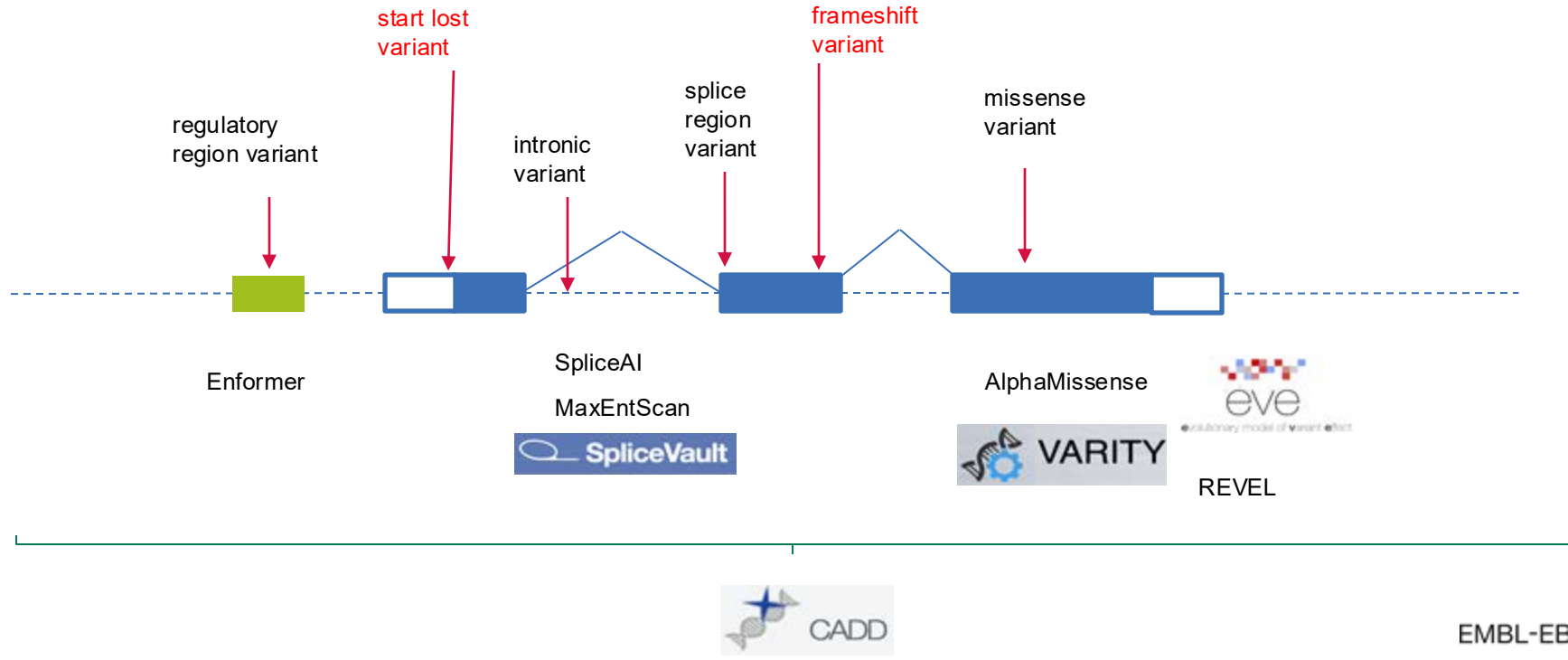


Variant scoring tools

- How it works
- Example command line

Variant scoring methods

In silico scoring of likely variant pathogenicity is an active area of research and different tools focus on different variant or function types



Variant scoring methods

- There are many computational methods to predict if a missense variant is likely to be deleterious
- Performance depends on data available and varies by gene/ region
- Some use protein homology information (eg. SIFT, PolyPhen-2, LRT)
- Some use the same training data (eg. FATHMM, MutationTaster, VEST3 use HGMD)
- Some incorporate other callers (eg.DANN, REVEL)
- Some use deep sequencing across species (eg. PrimateAI, AlphaMissense)

=> Agreement on pathogenicity rating may be due to similarity of method or noise in shared input/training data

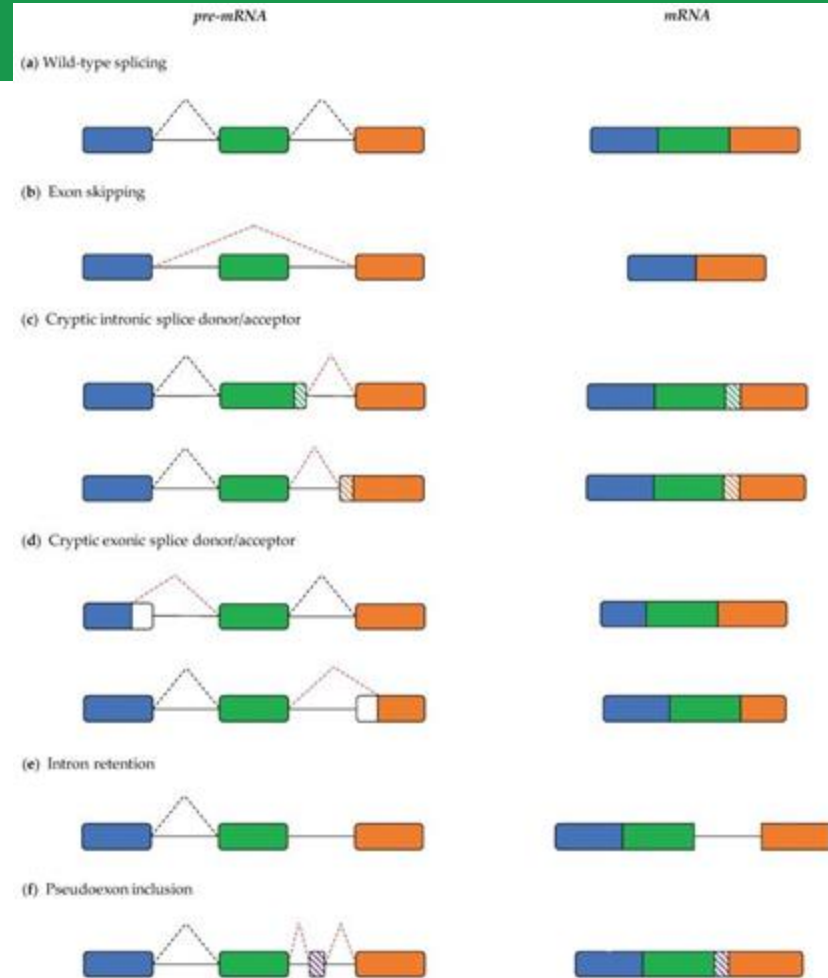
Variant impact on splicing

Genomic changes disrupting normal splice sites or creating novel splice sites can impact transcript structure and protein function

Ensembl VEP highlights when variants lie in regions important for splicing

Multiple tools exist to predict the disruption of existing or creation of novel (cryptic) splice sites

Image from Rowlands et al 2019
Machine Learning Approaches for the Prioritization
of Genomic Variants Impacting Pre-mRNA Splicing.



Splicing predictors

SpliceAI

- Deep learning method
- Predicts variant gain or loss of acceptor and donor (Δ score ≥ 0.8 confidently considered deleterious)
- Ensembl precalculated scores
 - calculated for all possible SNV for MANE transcripts
 - maximum distance between the variant and gained/lost splice site: 50bp

Also available in Ensembl VEP

- SpliceVault - precalculated predictions of exon-skipping events and activated cryptic splice sites based on the most common mis-splicing events around a splice site
- MaxEntScan - calculates on the fly

Outline

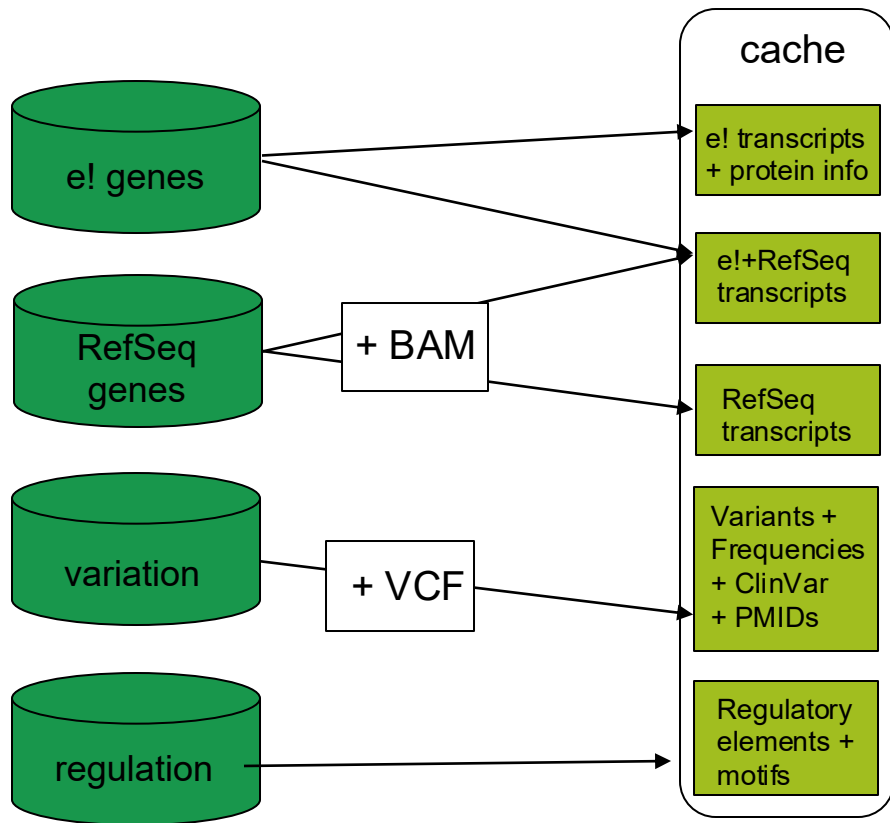
- Using Ensembl VEP
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- Variant scoring tools



How it works

- Example command line

Ensembl VEP data caches



Designed for quick reference data retrieval

Transcript caches

- Contain serialised Ensembl transcript objects
- Use NCBI alignments to RefSeq transcripts
- Created for Ensembl, RefSeq or both

Variant caches

- Simple tabix'ed files with ordered data

Regulation caches

- Contain serialised Ensembl regulatory feature objects

Info file – variation data available

Synonym file – reference sequence names

Updated with the latest data each Ensembl release

Extending Ensembl VEP - custom data sources

The Ensembl VEP API can extract reference data from standard bioinformatics file formats to use in annotation:

- Transcript annotation in GFF3 or GTF
- Variant frequencies, other INFO from VCF
- Genomic features in BED

Example use cases:

- Additional/new variant frequency sets such as AllofUs
- Specific gnomAD panels
- Private data sets
- MAVE scores associated with genomic coordinates & single base changes



http://www.ensembl.org/info/docs/tools/vep/script/vep_custom.html

Extending Ensembl VEP - Plugins

- Discrete modules to add specific functionality
- Optionally added to the Ensembl VEP command line
- Can be written quickly when a new tool or data set is available
- Common actions:
 - Run additional algorithm
 - Look up and manipulate pre-computed values from a file
 - Look up additional data via the Ensembl API

```
package FunkyPlugin;

use base qw(Bio::Ensembl::Variation::Utils::BaseVepPlugin);

sub feature_types {
    return ['Transcript'];
}

sub get_header_info {
    return {
        FUNKY_PLUGIN => "Description of funky plugin"
    };
}

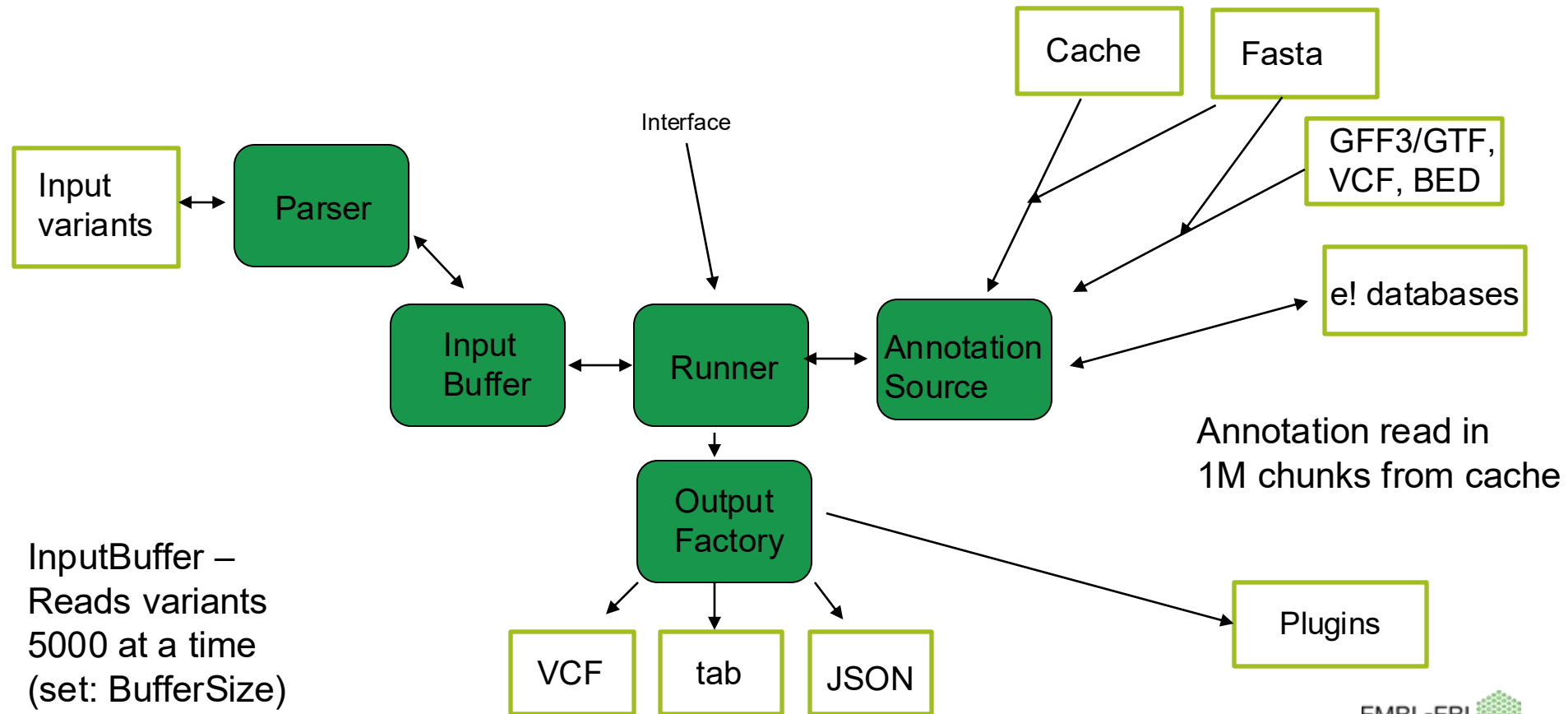
sub run {
    my ($self, $transcript_variation_allele) = @_;

    my $results = ... # do analysis

    return {
        FUNKY_PLUGIN => $results
    };
}

1;
```

Ensembl VEP process



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Example command line

Using command line Ensembl VEP

```
./vep --input_file [input_data] --output_file [output_file] --cache --fasta Homo_sapiens.GRCh38.fa.gz
```

file of variants
to annotate

define a file to
write results to

use Ensembl VEP
cache to look up
reference data

supply genomic
sequence for
quick look up

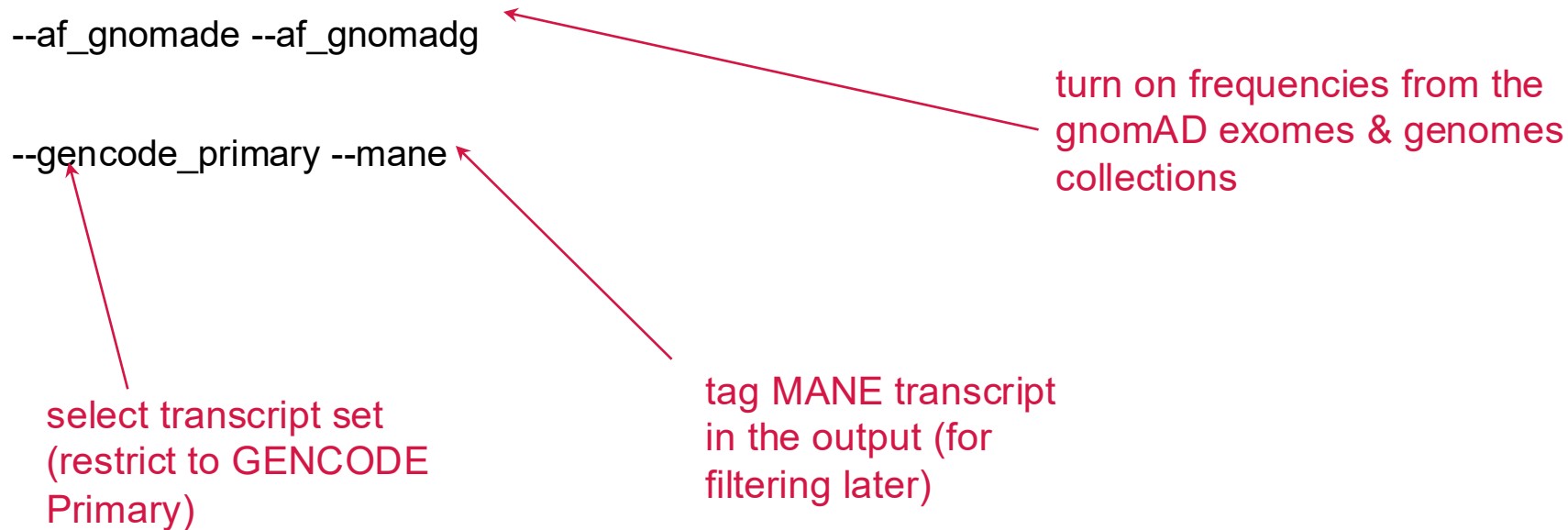
Using command line Ensembl VEP

```
./vep --input_file [input_data] --output_file [output_file] --cache --fasta Homo_sapiens.GRCh38.fa.gz
```

```
--af_gnomade --af_gnomadg
```

```
--gencode_primary --mane
```

turn on frequencies from the
gnomAD exomes & genomes
collections



select transcript set
(restrict to GENCODE
Primary)

tag MANE transcript
in the output (for
filtering later)

Using command line Ensembl VEP

```
./vep --input_file [input_data] --output_file [output_file] --cache --fasta Homo_sapiens.GRCh38.fa.gz
```

```
--af_gnomade --af_gnomadg
```

```
--gencode_primary --mane
```

```
--plugin AlphaMissense,file=/full/path/to/file.tsv.gz
```

```
--plugin REVEL,file=[path_to]/new_tabbed_revel_grch38.tsv.gz
```

```
--plugin SpliceAI,snv=[path_to]/spliceai_scores.raw.snv.ensembl_mane.grch38.110.vcf.gz
```

include plugins to
integrate precalculated
results from variant
scoring tools



Using command line Ensembl VEP

```
./vep --input_file [input_data] --output_file [output_file] --cache --fasta Homo_sapiens.GRCh38.fa.gz
```

```
--af_gnomade --af_gnomadg
```

```
--gencode_primary --mane
```

```
--plugin AlphaMissense,file=/full/path/to/file.tsv.gz
```

```
--plugin REVEL,file=[path_to]/new_tabbed_revel_grch38.tsv.gz
```

```
--plugin SpliceAI,snv=[path_to]/spliceai_scores.raw.snv.ensembl_mane.grch38.110.vcf.gz
```

```
--plugin IntAct,mapping_file=[path_to]/mutation_map.txt.gz,mutation_file=[path_to]/mutations.tsv,all=1
```

Add a plugin to highlight sites known to be important in interactions



Using command line Ensembl VEP

```
./vep --input_file [input_data] --output_file [output_file] --cache --fasta Homo_sapiens.GRCh38.fa.gz
```

```
--af_gnomade --af_gnomadg
```

```
--gencode_primary --mane
```

```
--plugin AlphaMissense,file=/full/path/to/file.tsv.gz
```

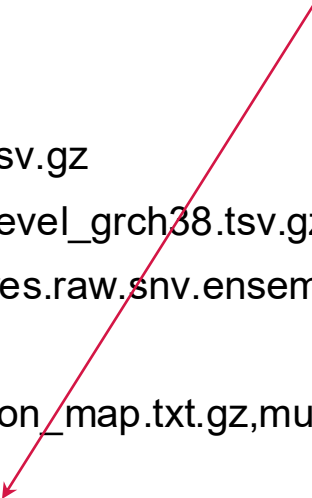
```
--plugin REVEL,file=[path_to]/new_tabbed_revel_grch38.tsv.gz
```

```
--plugin SpliceAI,snv=[path_to]/spliceai_scores.raw.snv.ensembl_mane.grch38.110.vcf.gz
```

```
--plugin IntAct,mapping_file=[path_to]/mutation_map.txt.gz,mutation_file=[path_to]/mutations.tsv,all=1
```

```
--plugin Conservation,/path/to/bigwigfile.bw
```

Add a plugin to report
conservation scores



Summary

There are increasing numbers of datasets and tools to aid in the assessment of likely variant impact

- they have a variety of formats & are regularly updated

Ensembl VEP simplifies the application of these resources

- It has bundled data caches
- It is simple to extend and configure



See MaveDB plugin for the annotation of human variants with results from MAVE assays
Arbesfeld et al 2025 doi:10.1186/s13059-025-03647-x

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